Project 3 - DataSEC 2026: Environmental Sound Classification under Acoustic Noise

Dataset & Problem

DataSEC is a curated benchmark collection of isolated environmental sound events published in *Scientific Data* (Nature Springer) in 2025 [1], designed for training and evaluating Sound Event Classification (SEC) models:

Property	Value
Clips	4,292 monophonic audio files in Waveform Audio File Format (.wav)
Total duration	Approximately 18 hours 26 minutes
Classes	22 macro-classes, 28 subclasses
Sampling rate	44.1 kilohertz (kHz)
Audio format	Monophonic, uncompressed linear Pulse-Code Modulation (PCM)

Classification vs. Detection: In Sound Event Classification (SEC), each audio recording contains a single isolated acoustic event belonging to a target category, and the model predicts which category is present. This is distinct from Sound Event Detection (SED), which requires estimating both the event category and its precise temporal onset and offset timestamps within continuous, multi-source audio streams.

Task: 22-class environmental sound classification operating on log-frequency Mel-scale spectrograms (log-Mel spectrograms). Students are not required to construct a continuous sound-event-detection system.

Computing constraint: All required experiments are designed to run on the free tier of Google Colab using a single NVIDIA T4 Graphics Processing Unit (GPU) with 16 GB Video RAM (VRAM). Use the exact pinned dataset version specified below.

Resource	Link
Original paper	https://doi.org/10.1038/s41597-025-05991-w
DataSEC (course version)	https://zenodo.org/records/17033970
DataSEC DOI	https://doi.org/10.5281/zenodo.17033970

Pin the dataset version. Employ Digital Object Identifier (DOI) `10.5281/zenodo.17033970` exclusively.

Requirements

Midterm

A. Dataset Analysis

Report:

- total number of audio clips and distribution across the 22 macro-classes;
- duration statistics per category (minimum, maximum, mean, median);
- time-domain waveforms and linear-frequency spectrograms for at least three contrasting sound categories;
- 2D log-Mel spectrogram representations;
- class balance assessment;
- identification of acoustic classes that exhibit overlapping spectral envelopes or frequent mutual confusion.

Explain the mathematical and physical distinctions among:

1. raw audio waveform (time-domain acoustic pressure oscillations);
2. linear-frequency spectrogram (magnitude squared of the Short-Time Fourier Transform, STFT);
3. log-Mel spectrogram (frequencies warped onto the non-linear human auditory Mel scale using triangular filterbanks, followed by logarithmic dynamic range compression).

B. Runtime Feature Pipeline

Implement a fully reproducible runtime pipeline - do not generate or persist pre-computed spectrogram image files on disk:

```
raw audio file (.wav)
  -> load and resample
  -> crop / pad to fixed duration (4-5 s)
  -> compute log-Mel spectrogram (64 or 128 Mel bins)
  -> neural network input
```

Recommended default parameters: 4-5 s fixed duration; 64 or 128 Mel frequency bins; sampling rate resampled to 16 kHz or 22.05 kHz.

C. Required Models

Train and evaluate:

1. Small Convolutional Neural Network (CNN) operating on 2D log-Mel spectrogram inputs;
2. MobileNetV3-Small [6] with pretrained ImageNet weights (single spectrogram channel replicated across 3 RGB input channels);
3. Convolutional Recurrent Neural Network (CRNN) [7]: CNN spatial feature extractor followed by a Gated Recurrent Unit (GRU) [8] temporal sequence encoder.

D. Required Metrics

- Top-1 Accuracy, Macro-averaged F1-score (Macro-F1);
- per-class F1-score;
- Confusion Matrix;
- model parameter count and wall-clock training time;
- computational complexity measured via Multiply-Accumulate Operations (MACs) / Floating-Point Operations (FLOPs) or inference latency (milliseconds per audio clip).

E. Midterm Questions

1. Which environmental sound classes achieve the highest and lowest recognition accuracy?
2. Which sound categories exhibit the most severe mutual confusion?
3. Does the recurrent temporal modeling in CRNN yield measurable improvements over pure 2D CNN feature extraction on varying-duration sounds?
4. Which architecture achieves the most favorable performance-compute trade-off?

F. Midterm Deliverables

- presentation slides;
- executable Jupyter notebook / Python codebase;
- dataset-analysis summary figures and tabular metrics;
- baseline experimental results table;
- confusion matrix and qualitative audio error audit;
- proposed research question for the Final evaluation;
- project README documentation.

Final

Required Research Question (RQ1):

How robust are environmental sound classifiers when the target sound is perturbed by realistic acoustic noise, temporal distortions, and spectral corruptions?

Utilize the clean test set established during the Midterm. Keep all test ground-truth labels unchanged. Apply controlled acoustic corruptions during test-time evaluation only.

A. Required Perturbations

Evaluate models under at least three of the following audio perturbations, each tested across three severity levels:

- additive background noise at varying Signal-to-Noise Ratio (SNR) levels (e.g., +20 dB, +10 dB, 0 dB SNR);
- audio gain / volume scaling (-12 dB, -6 dB, +6 dB);
- circular time shifting;
- temporal masking (zeroing out random time intervals);
- frequency masking (zeroing out selective spectral bands);
- controlled environmental background mixing (blending target sound with ambient street noise or rain soundscapes).

B. Required Comparison

Evaluate at least three model architectures. For each model, report:

$$F_1^{clean}, \quad F_1^{corrupted}, \quad \Delta F_1 = F_1^{clean} - F_1^{corrupted}.$$

Plot performance degradation curves across corruption severity levels for each perturbation type.

C. Required Error Analysis

- per-class degradation under each corruption type;
- confusion-matrix transitions (clean test set versus corrupted test set);
- concrete audio examples where models flip from correct classification to incorrect predictions;
- qualitative discussion linking acoustic properties (e.g., harmonic structure, transient impulses, broadband energy) to observed failure modes.

D. Final Deliverables

- conference-style technical report;
- reproducible Jupyter notebook / Python codebase;
- fixed split index files;
- configuration files documenting feature extraction parameters;
- saved prediction probabilities per model and condition (CSV file with columns: `sample_id`, `true_class`, `pred_class`, `probabilities`, `duration`, `model`, `split`, `perturbation`, `severity`);
- programmatic plots and comparison tables;
- project README with reproduction instructions.

Optional Extensions Attempt these extensions only after RQ1 has been fully executed and validated.

Extension 1 - SpecAugment Data Augmentation (RQ2)

Evaluate whether incorporating SpecAugment [9] (random time masking and frequency masking applied directly to spectrograms during training) attenuates the clean-to-corrupted generalization drop under acoustic perturbations [2], [5].

Extension 2 - Hierarchical Taxonomic Classification (RQ3)

Exploit the two-level taxonomy of DataSEC (22 macro-classes and 28 subclasses) to compare flat 22-class classification against a hierarchical two-stage or multi-task prediction framework.

Extension 3 - Accuracy-Efficiency Pareto Analysis (RQ4)

Benchmark Macro-F1 against parameter count, inference latency, wall-clock training time, and peak GPU Video RAM (VRAM) consumption. Identify Pareto-efficient architectures across both clean and acoustically corrupted conditions.

Extension 4 - Transfer from Isolated Events (DataSEC) to Continuous Soundscapes (DataSED) (RQ5, paper track)

Evaluate models trained on isolated DataSEC events against continuous real-world acoustic recordings from the companion DataSED dataset [1], [10] using ground-truth event boundaries:

1. exact ground-truth event temporal boundaries;
2. event segment accompanied by short surrounding temporal context (+/- 1 second);
3. fixed-length contextual windows (4-5 seconds centered on the target event).

This protocol represents **event classification with oracle temporal boundaries** - not end-to-end sound-event detection. The central scientific inquiry is whether neural representations learned from isolated, curated events generalize to identical acoustic event categories embedded in unconstrained, continuous soundscapes [3], [4].

External companion dataset: DataSED, DOI `10.5281/zenodo.15346092` [10].

Reproducibility Rules

- Pin the exact dataset version (Zenodo DOI `10.5281/zenodo.17033970`).
- Fix train / validation / test partition IDs.
- Never adjust data augmentation parameters based on test-set performance feedback.
- Save continuous probability distributions for every model and experimental condition.
- Document audio sampling rate, Short-Time Fourier Transform (STFT) window parameters, random seeds, and Python dependency versions.
- For paper-track submissions, run experiments across at least three distinct random seeds.

References

[1] L. Fredianelli, F. Artuso, G. Pompei, G. Licitra, G. Iannace, and A. Akbaba, "Environmental Noise Dataset for Sound Event Classification and Detection," *Scientific Data*, vol. 12, no. 1, Oct. 2025, doi: 10.1038/s41597-025-05991-w.

[2] R. Akter, Md. R. Islam, S. K. Debnath, P. K. Sarker, and Md. K. Uddin, "A hybrid CNN-LSTM model for environmental sound classification: Leveraging feature engineering and transfer learning," *Digital Signal Processing*, vol. 163, p. 105234, Aug. 2025, doi: 10.1016/j.dsp.2025.105234.

[3] B. Chen and J. Peng, "Environmental sound classification using two-stream deep neural network with interactive time-frequency attention," *Applied Acoustics*, vol. 238, p. 110794, Sep. 2025, doi: 10.1016/j.apacoust.2025.110794.

[4] N. Tonami et al., "Trainingless Adaptation of Pretrained Models for Environmental Sound Classification," in *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, Apr. 2025, pp. 1-5, doi: 10.1109/icassp49660.2025.10889784.

[5] H. Xu, H. Chen, X. Liu, and H. Ren, "Improving one-dimensional-based environmental sound classification models with progressive projection knowledge distillation from two-dimensional-based models," *Engineering Applications of Artificial Intelligence*, vol. 151, p. 110622, Jul. 2025, doi: 10.1016/j.engappai.2025.110622.

[6] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 1314-1324, doi: 10.1109/ICCV.2019.00140.

[7] K. Choi, G. Fazekas, M. Sandler, and K. Cho, "Convolutional recurrent neural networks for music classification," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP)*, 2017, pp. 2392-2396,

doi: 10.1109/ICASSP.2017.7952585.

[8] K. Cho et al., "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2014, pp. 1724-1734, doi: 10.3115/v1/D14-1179.

[9] D. S. Park et al., "SpecAugment: A simple data augmentation method for automatic speech recognition," in *Proc. Interspeech*, 2019, pp. 2613-2617, doi: 10.21437/Interspeech.2019-2680.

[10] L. Fredianelli et al., "DataSED: Curated environmental sound dataset for sound event detection," Zenodo, 2025, doi: 10.5281/zenodo.15346092.